

Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Kinematics 7 Extension

Total _____ / 23

A. Which equation? $a = \frac{v-u}{t}$ or $v^2 - u^2 = 2as$ — the empty line in your Givens box tells you.

1) **Calculate** the speed a go-kart was moving at before it accelerated at 2.5 m/s^2 for 4.0 s , reaching 18 m/s . [3]

Givens

$s =$

$u =$

$v =$

$a =$

$t =$

2) **Calculate** the acceleration of a motorbike that brakes from 24 m/s to a stop in 48 m . [3]

Givens

$s =$

$u =$

$v =$

$a =$

$t =$

3) **Calculate** the speed of a hockey puck after it slides 40 m across the ice. It is hit at 5.0 m/s and slows down with an acceleration of -0.20 m/s^2 . [3]

Givens

$s =$

$u =$

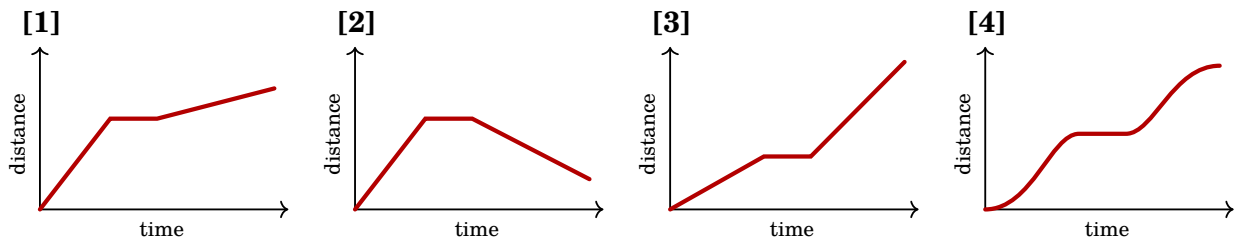
$v =$

$a =$

$t =$

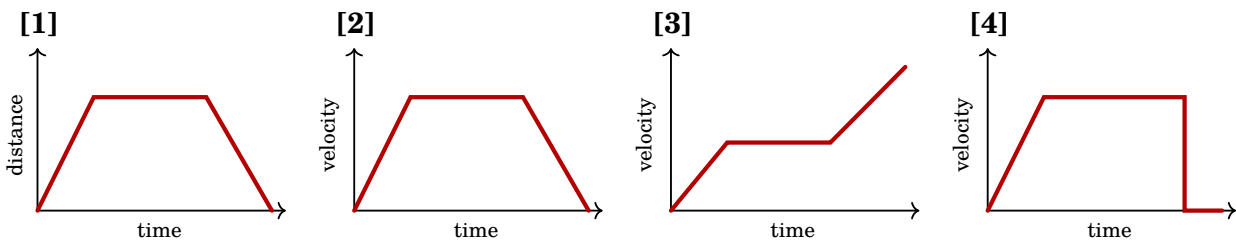
B. Graphs *choose the graph that matches.*

- 4) A bus drives at a steady speed, stops at a bus stop, then drives on **faster**. Which distance–time graph shows this journey? [2]



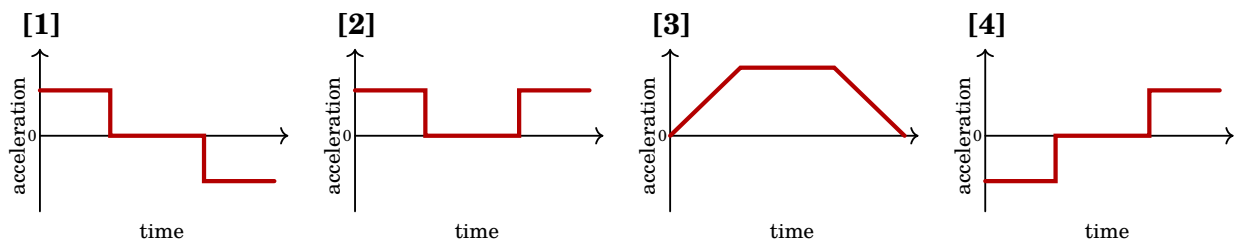
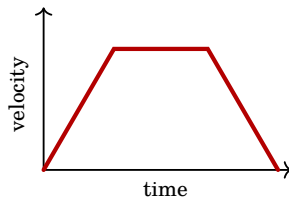
Graph _____ because

- 5) A lift speeds up steadily from rest, moves at a constant speed, then slows steadily to a stop. Which graph shows this? [1]



Graph _____

- 6) Here is a velocity–time graph. Which acceleration–time graph matches it? [1]



Graph _____

- 7) A motor boat leaves a jetty, then slows down to moor. The graph shows its motion.



- (a) Between which times is the acceleration greatest? Explain how the graph shows this.

[2]

- (b) Calculate the acceleration between 2 s and 6 s.

[1]

- (c) Describe the motion of the boat from 6 s to 13 s.

[2]

- (d) Use $v^2 - u^2 = 2as$ and your answer to (b) to calculate the distance travelled between 2 s and 6 s.

Givens

$s =$

$u =$

$v =$

$a =$

$t =$

[2]

- (e) Use the area under the graph to calculate the total distance the boat travels. The area between 2 s and 6 s matches your answer to (d).

[3]